

Therese Joseph

Lead Research Engineer

Brisbane, Australia

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EDUCATION

Queensland University of Technology (QUT)

Doctor of Philosophy (PhD) in Robotics

Brisbane, Australia

Mar 2022 – Dec 20

- Thesis: *Heterogeneous Sensing Methodologies for Robust Localisation in Challenging Conditions*

Queensland University of Technology (QUT)

Bachelor of Engineering (Mechatronics), Minor in Robotics & Adv. Electrical

Brisbane, Australia

2018 – 2021

- GPA: 6.72 / 7.0
- Dean's List Award for High Academic Achievement: 2018, 2019, 2020, 2021

EXPERIENCE

Local Positioning Systems (Startup)

Lead Research Engineer

Brisbane, Australia

Jan 2026 – Present

- Leading the iMOVE-funded project: "The Trusted Transport Positioning System".
- Developing a deployable LPS hardware kit and software backend for tracking scooters and on-road vehicles.
- Creating GUI visualisations for mapping transport environments and live position tracking in GNSS-denied areas.
- Conducting performance evaluations using standard positioning metrics across diverse road environments (e.g., tunnels, inner cities).
- Managing assets, equipment procurement, safety compliance, and quarterly deliverables.

QUT Centre for Robotics

Research Assistant

Brisbane, Australia

Oct 2024 – Dec 2024

- Developed a custom visualiser using the Google Maps API to display position information and mapped regions.

Queensland University of Technology (QUT)

PhD Candidate

Brisbane, Australia

Mar 2022 – Dec 2025

- Specialising in robust localisation using LiDAR in unstructured forests and Event Cameras for challenging illumination conditions.
- Developing algorithms based on bio-inspired networks for challenging visual conditions.

Queensland University of Technology (QUT)

University Tutor

Brisbane, Australia

July 2022 – Nov 2024

- Tutored a third year and honours telecommunication courses in Electrical Engineering (5 semesters total).
- Roles included presenting content to a class of up to 30 students, answering questions on online forums, and marking assignments.

CSIRO's Data61

Research Vacation Program / Honours Student

Pullenvale, QLD

June 2020 – Dec 2021

- Assisted in research projects involving literature review, design, and experimentation.
- Modeled and 3D printed innovative designs for soft robotics and conducted tests to validate their performance.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, OpenCV), Swift, Kotlin, MATLAB

Robotics: ROS, vSLAM, LiDAR, Event Cameras, Soft Robotics

Development Tools: Git, Linux, LaTeX

Hardware & Design: CAD (Solidworks), 3D Printing (Soft & Hard materials)

PUBLICATIONS

1. **T. Joseph**, T. Fischer, M. Milford. “Ensemble-Based Event Camera Place Recognition Under Varying Illumination.” *IEEE Robotics and Automation Letters (RA-L)*, 2026. Also accepted to be presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
2. **T. Joseph**, T. Fischer, M. Milford. “Matched Filtering based LiDAR Place Recognition for Urban and Natural Environments.” *IEEE Robotics and Automation Letters (RA-L)*, 2025. Also accepted to be presented at the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
3. **T. Joseph**, T. Fischer, M. Milford. “Trajectory Tracking via Multiscale Continuous Attractor Networks.” In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
4. **T. Joseph**, S. Baldwin, L. Guan, J. Brett, D. Howard. “The jamming donut: a free-space gripper based on granular jamming.” *IEEE International Conference on Soft Robotics (RoboSoft)*, 2023.
5. D. Howard, J. O’Connor, J. Letchford, **T. Joseph**, et al. “A comprehensive dataset of grains for granular jamming in soft robotics: Grip strength and shock absorption.” *IEEE International Conference on Soft Robotics (RoboSoft)*, 2023.
6. D. Howard, J. O’Connor, **T. Joseph**, et al. “Getting a grip: In materio evolution of membrane morphology for soft robotic jamming grippers.” *IEEE RoboSoft*, 2022.